Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION – APRIL 2021

ASCS402Q - STATISTICAL METHODS FOR COMPUTER APPLICATIONS – II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2=20)

Answer ALL the questions

1. Define independent events.
2. Define Statistical Probability.
3. State the properties of distribution function.
4. Define probability mass function.
5. State the recurrence relation formula for probabilities of Poisson distribution.
6. What are the conditions for normal distribution as a limiting form of Binomial distribution?
7. Define Student's t-statistic.
8. Define Standard error.
9. Define randomization.
10. What is local control?

SECTION - B (5×5=25)

Answer ALL the questions

11. a. If two dice are thrown. What is the probability that the sum is
    i. Greater than 8
    ii. neither 7 nor 11.

    (Or)

    b. A problem in Statistics is given to three students A, B and C whose chances of solving it are 1/2 , 3/4 and 1/4 respectively. What is the probability that the problem will be solved if all of them try independently.

12. a. Explain the properties of expectation.

    (Or)

b. An urn contains 7 white and 3 red balls. Two balls are drawn together at random from this urn. Compute the probability that neither of them is white. Find also the probability of getting one white and one red ball. Hence compute the expected number of white balls drawn.

13. a. With the usual notations, find p for a binomial variate X, if n=6 and

9 P(X= 4)= P(X = 2).

(Or)

b. The marks obtained by a number of students for a certain subject are assumed to be approximately normally distributed with mean value 65 and with a standard deviation of 5 If 3 students are taken at random from this set. What is the probability that exactly 2 of them will have marks over 70?

14. a. It is believed that the precision (as measured by the variance) of an instrument is no more than 0.16. Carry out the test at 1% level given 11 measurements of the same subject on the instrument.

2.5 2.3 2.4 2.3 2.5 2.7 2.5 2.6 2.6 2.7 2.5

(Or)

b. A random sample of 16 values from a normal population showed a mean of 41.5 inches and the sum of squares of deviations from this mean equal to 135 square inches. Show that the assumption of a mean of 43.5 inches for the population is not reasonable.

15. A. Explain the one way classification.

(Or)

b. Explain the following terms.

i. Experiment
ii. Treatments
iii. Experimental Error
iv. Blocks
v. Yield

SECTION - C (3×10=30)

Answer any THREE Questions

16. State and prove Baye's Theorem.

17. State and prove Chebyshev's inequality.

18. Fit a Poisson distribution to the following data.

X:	0	1	2	3	4
F:	109	65	22	3	1

19. In a year there are 956 births in a town A of which 52.5% were males, while in towns A and B combined, this proportion in a total of 1406 births was 0.496. Is there any significant difference in the proportion of male births in the two towns?

20. Analyse the following Randomized Block Design.

	Blocks			
Treatments	B1	B2	B3	B4
T1:	5	8	10	12
T2:	10	12	10	8
T3:	16	10	14	12
T4:	8	6	10	12

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